

Detailed Proposal: 3D Escape Room Game

Student Name: Pengyu Lu

Student ID: 201822113

Statement of Ethical Compliance

This project is categorised as A2 for human participants and A for data. Human participants will be involved only in software evaluation activities, which falls under the department's collective ethics approval. The following ethical guidelines will be strictly followed:

- All participation will be voluntary, with participants able to withdraw at any time.
- No personal or identifiable data will be collected during evaluation sessions.
- Participants will be provided with information sheets and asked to sign consent forms.
- All evaluation data will be aggregated and anonymised before analysis.
- All participant data will be stored securely and deleted after graduation.

Project Description

This project aims to design and develop an interactive 3D escape room game using the Unity game engine. Unlike traditional point-and-click escape games, which rely on static images and pre-defined hotspots, this game will allow players to move freely in a 3D environment, interact with objects in real-time, and solve puzzles through exploration and logical reasoning. The game will feature one or more themed rooms, each containing a series of puzzles that must be solved to progress. The goal is to create an engaging and immersive experience that demonstrates the potential of 3D environments in the escape room genre.

Aims & Requirements

Aims:

- To design and develop a fully interactive 3D escape room game using Unity.
- To implement intuitive player controls and a user-friendly interface.
- To create logical and engaging puzzles that encourage exploration and problem-solving.
- To evaluate the gameplay experience through user testing and feedback.
- (Stretch Aim) To develop a reusable game engine or API that allows others to create their own 3D escape rooms.

Requirements:

Essential:

- First-person player movement (walk, look, interact)
- Interactive objects (e.g., pick up, examine, combine)
- Multiple puzzles per room with logical solutions
- Clear win/lose conditions
- Basic UI for inventory and instructions

Desirable:

- Multiple themed rooms with unique puzzles
- Sound effects and background music
- Save/load game functionality
- Modular room design for easy expansion
- Exportable game engine/API for custom escape rooms

Key Literature & Background Reading

The project draws inspiration from both classic escape room games and modern 3D adventure games. Key references include:

- Unity Technologies. (2023). *Unity User Manual*. [Online] Available at: <https://docs.unity3d.com/Manual/index.html>
This will be the primary technical reference for implementing game mechanics.
- Johnson, M. (2021). *Game Design Principles for Puzzle Games*. Game Developer.
This article discusses puzzle design and player engagement.
- Adams, E. (2014). *Fundamentals of Game Design*. New Riders.
A foundational text on game design principles, including level design and interactivity.
- Unreal Engine. (2022). *Blueprint Visual Scripting*. [Online] Available at: <https://dev.epicgames.com/documentation/en-us/unreal-engine/introduction-to-blueprints-visual-scripting-in-unreal-engine>

Although this project uses Unity, Unreal's approach to visual scripting informs the design of a reusable engine.

- “The Room” series by Fireproof Games – an example of high-quality 3D puzzle games.

Development & Implementation Summary

Development Environment:

- Unity 2022 LTS (Stable operation)
- C# for scripting
- Blender for 3D modeling (Depends on models required)
- Git for version control

Implementation Approach:

The game will be built in modular components:

1. **Player Controller** – Handles movement, camera control, and interaction.
2. **Interaction System** – Manages object highlighting, pickup, and use.
3. **Puzzle Manager** – Tracks puzzle states and triggers progression events.
4. **Inventory System** – Allows players to collect and use items.
5. **UI System** – Displays inventory, hints, and game status.

Workflow will follow an iterative Agile approach, with weekly sprints focusing on specific features (e.g., movement, interaction, puzzle logic).

Data Sources

This project does not involve the use of external datasets or personal data. All 3D models and assets will be created from scratch or sourced from royalty-free asset stores (e.g., Unity Asset Store) with appropriate licenses. No personal or sensitive data will be collected or processed.

Testing & Evaluation

Testing:

- Unit tests for key scripts (e.g., inventory, puzzle logic)
- Playtesting to identify bugs and usability issues
- Performance testing to ensure smooth frame rates

Evaluation:

- User testing with peers to assess engagement and difficulty
- Feedback forms to collect qualitative data on gameplay experience
- Success will be measured by completion time and user satisfaction

Project Ethics & Human Participants

Human participants will be involved solely during the software evaluation phase.

Participants will be asked to:

- Playtest the game and provide feedback on engagement and usability
- Complete a brief questionnaire about their gameplay experience
- Participate in short discussions about game mechanics and puzzle design

All activities will follow the university's ethical guidelines:

- Participants will receive information sheets and sign consent forms
- No personal data will be collected; all feedback will be anonymised
- Screen recordings may be used with permission, but will focus only on gameplay
- Audio recordings will be transcribed and then deleted promptly
- All data will be aggregated for analysis in the dissertation

These activities are permitted under the department's collective approval for low-risk software evaluation.

BCS Project Criteria

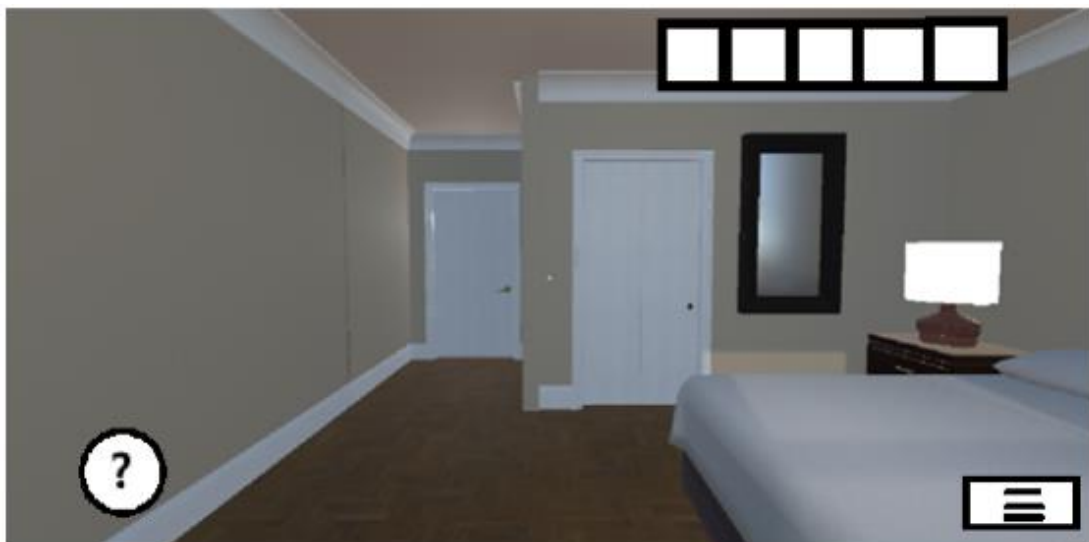
- **Practical & Analytical Skills:** Demonstrated through the use of Unity, C#, and game design principles.
- **Innovation/Creativity:** Moving escape rooms from 2D to 3D with free movement is an innovative step.
- **Synthesis & Evaluation:** The project combines game design, programming,

and user testing to deliver a polished product.

- **Real-World Need:** Escape rooms are popular in both digital and physical forms; this project explores new interaction models.
- **Self-Management:** The use of a Gantt chart and iterative development supports effective time management.
- **Critical Self-Evaluation:** Ongoing reflection and user feedback will inform improvements.

UI/UX Mockup

Below is a game's main screen:

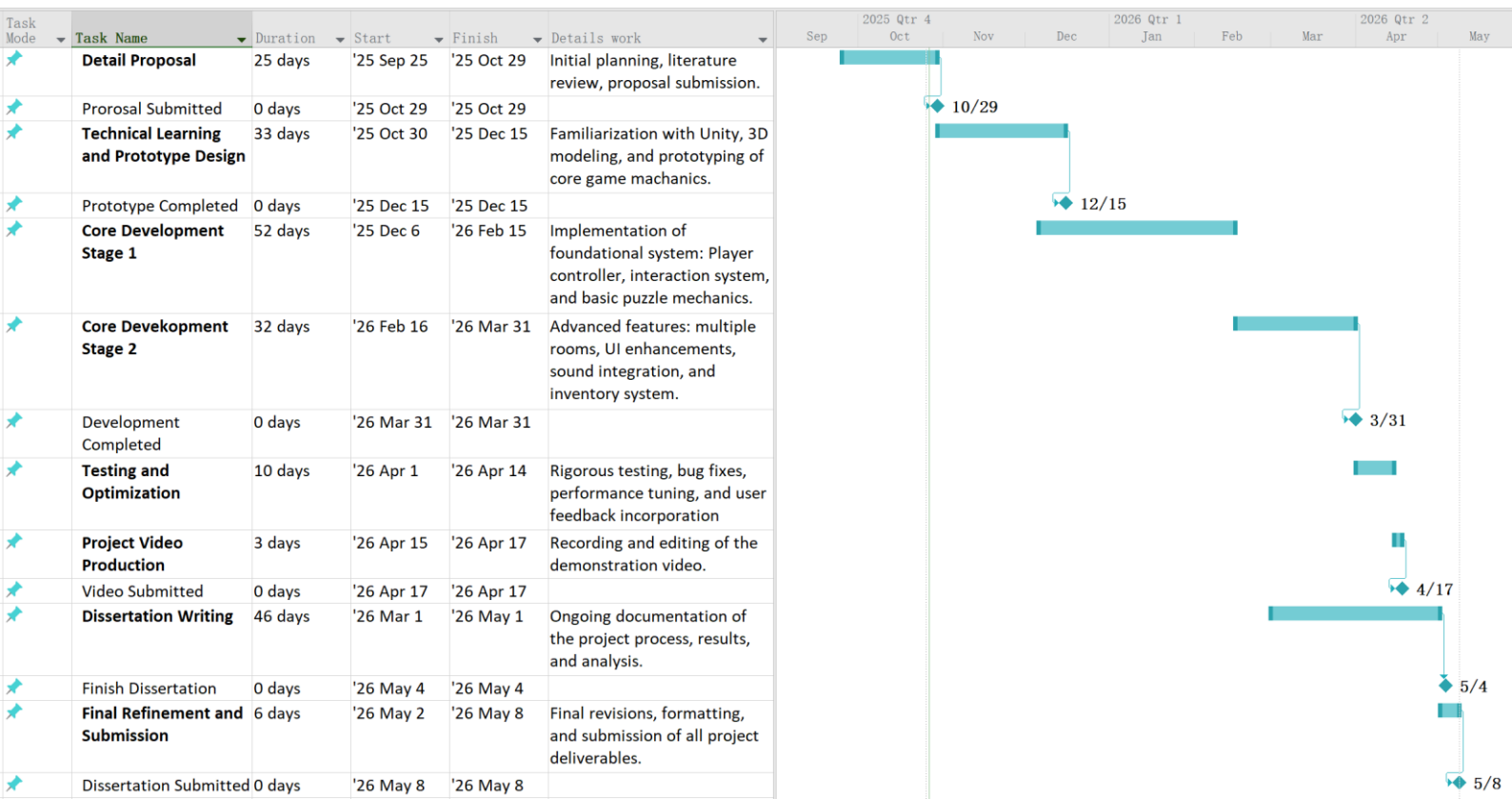


The game will use a first-person perspective and minimal HUD. The inventory column will appear in the upper right corner of the screen. When the player hovers the mouse over an interactive object, an interactive prompt will appear. The prompt button is located at the bottom left of the screen, and when the player clicks it, it will provide specific solutions to the problem. The menu button is located at the bottom right of the screen, and the player can return to the menu through this button.

Project Plan

The project follows a structured timeline from September 2025 to May 2026. The schedule is visualized in the Gantt chart below.

Gantt chart:



Risks & Contingency Plans

Risk	Contingency Plan	Likelihood	Impact
Hardware failure	Regular backups to cloud and external drives	Low	High
Scope creep	Prioritize core features; defer extras	Medium	Medium
Technical challenges	Seek supervisor guidance; use online resources	High	Medium
Time management	Follow Gantt chart; adjust scope if needed	Medium	High

References

- Unity Technologies. (2023). *Unity Documentation*. Available at: <https://docs.unity3d.com/Manual/index.html>
- Adams, E. (2014). *Fundamentals of Game Design*. New Riders. Available at: https://www.academia.edu/81247135/Fundamentals_of_game_design
- Fireproof Games. (2012). *The Room*. [Video Game].
- Johnson, M. (2021). *Game Design Principles for Puzzle Games*. Game Developer. Available at: <https://www.gamedeveloper.com/>
- Blender Foundation. (2023). *Blender*. Available at: <https://www.blender.org/>